# HDB Monthly Rent Prediction

## Progress Report

CS5228 Competition Project

Exploratory analysis and first validation evidence

**150,000 labeled records | 50,000 competition inputs**

Observed approval dates: 2021-01 to 2026-07

Local analysis of the supplied project data

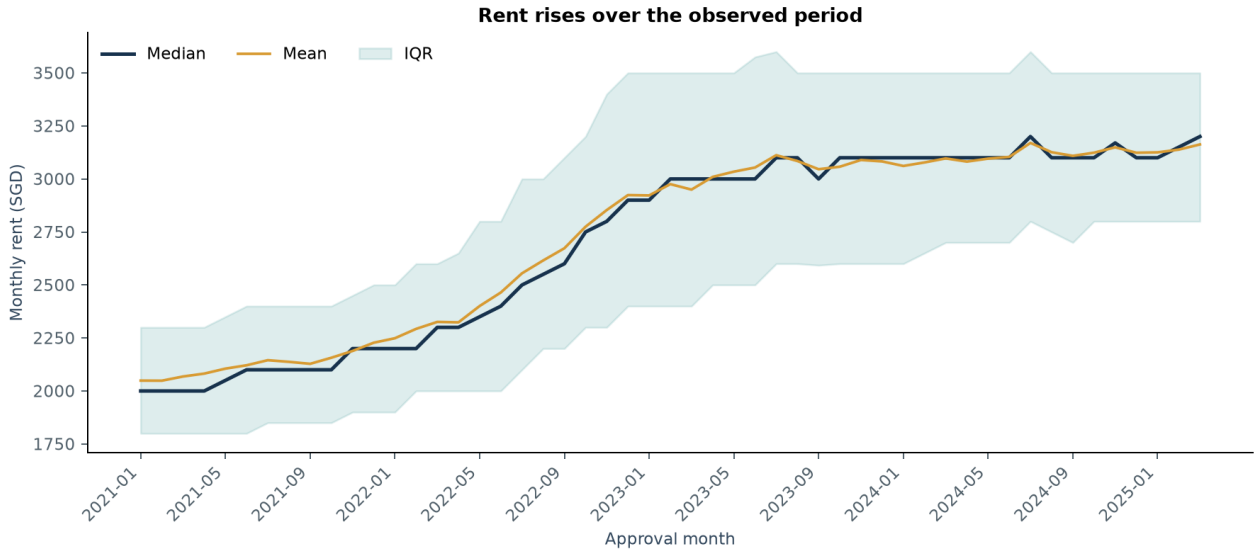
# Project question and analysis plan

## Question

How well can property, location and approval-time information predict monthly HDB rent for future records?

## Current design

- Audit the supplied data and label distributions.
- Compare random and chronological holdouts.
- Test property and location features before adding date-aware economic series.




Source: data/train.csv, data/test.csv; EDA protocol in analysis/eda/comprehensive.py

# Competition data and prediction boundary

| | | |
|---|---|---|
| **Labeled training** | **150,000 rows** | 2021-01 to 2025-03 |
| **Competition test** | **50,000 rows** | 2025-03 to 2026-07 |
| **Target** | **MONTHLY_RENT** | SGD per month |
| **Inputs** | **10 shared fields** | home, place, approval month |

The test window extends beyond the labeled period, so future-period validation is central to estimating competition performance.

Source: data/train.csv and data/test.csv; row counts and dates: reports/eda/summary.json

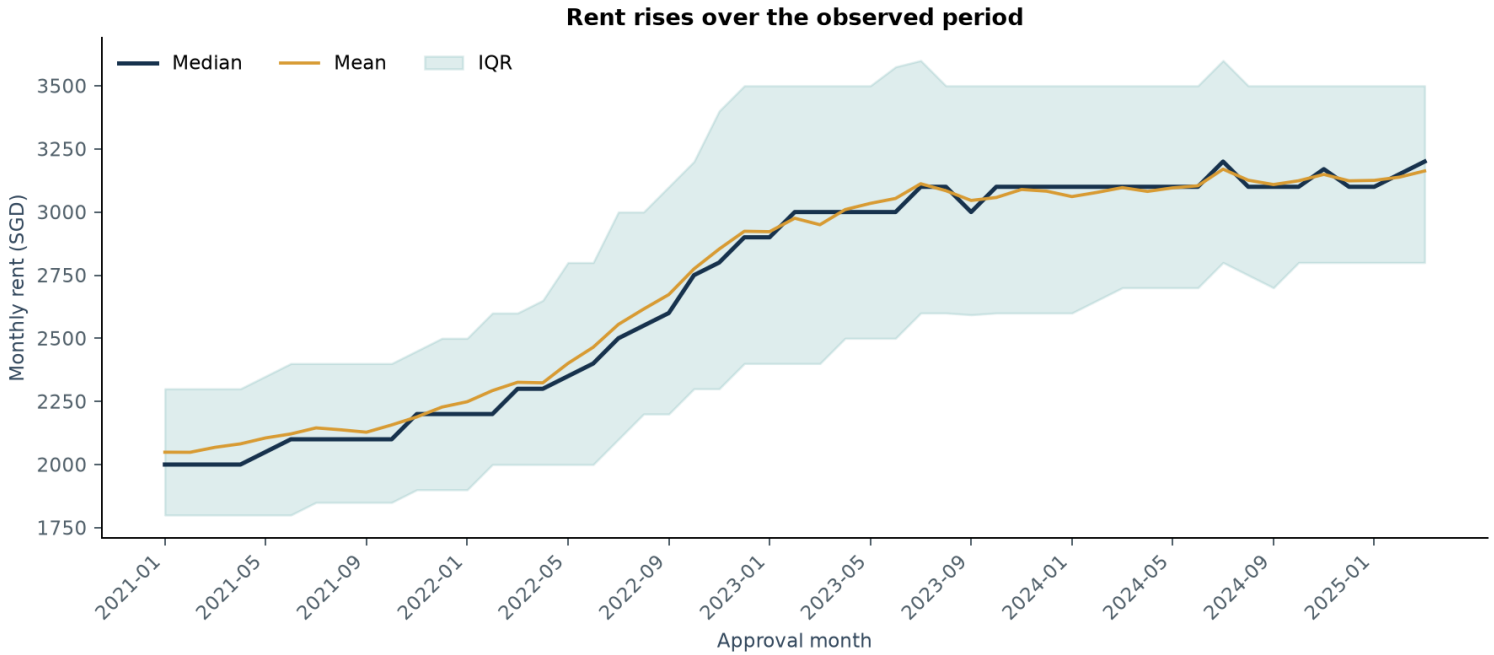
# Target distribution




- Mean $2,719; median $2,700 per month.
- Middle 50%: $2,100 to $3,200.
- Skewness 0.39; observed range $300 to $7,600.
- Mild right skew does not by itself justify a log target; compare validation errors before choosing a transform.

Source: reports/eda/summary.json; reports/eda/figures/01_target_distribution.png

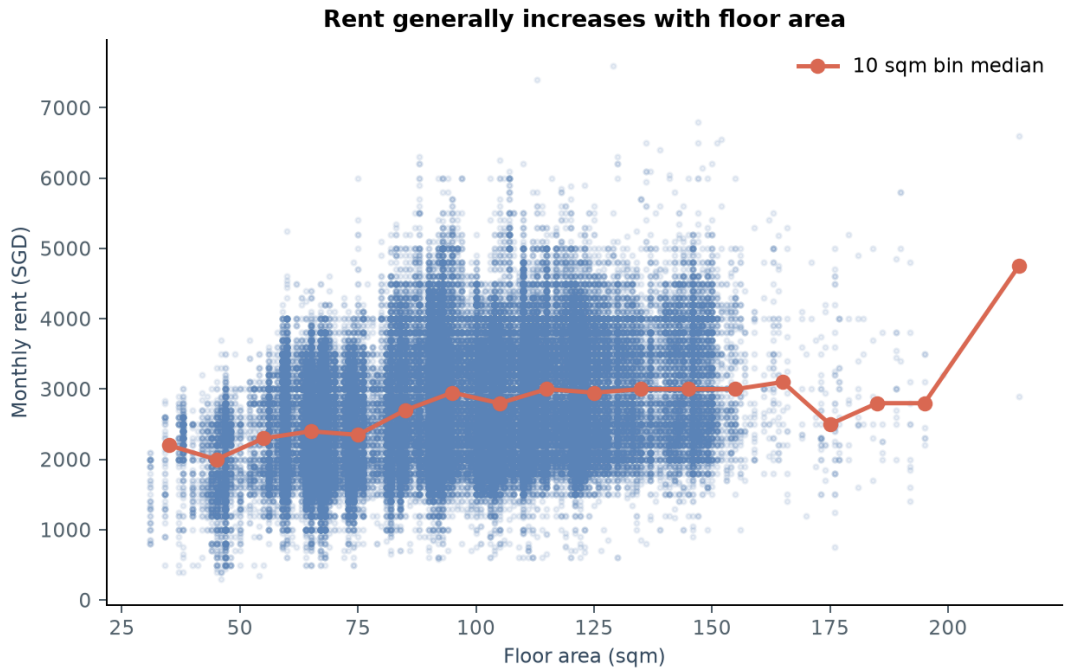
# Rent levels change across approval time



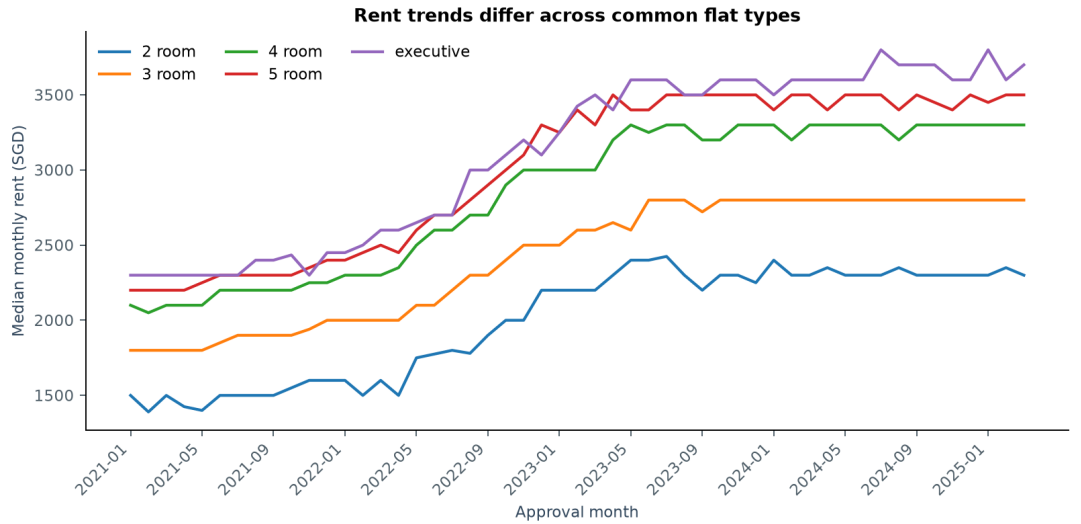

- Annual median: $2,100 in 2021 to $3,150 in 2025.
- The 2025 figure covers 8,924 rows through March only.
- Time is a first-order feature candidate and a reason to report chronological validation.

Source: reports/eda/tables/annual_rent_trend.csv; monthly series in reports/eda/tables/monthly_rent_trend.csv

# Property and flat-type patterns




Area spans 31-215 sqm (median 92); rank association with rent is 0.32. Large-area bins are sparse.




Flat-type rent trends rise over time, with different levels by type. Normalize spelling before encoding categories.

Source: reports/eda/figures/05_rent_vs_floor_area.png and 09_monthly_rent_by_flat_type.png

# Block reference data provides full coordinate coverage

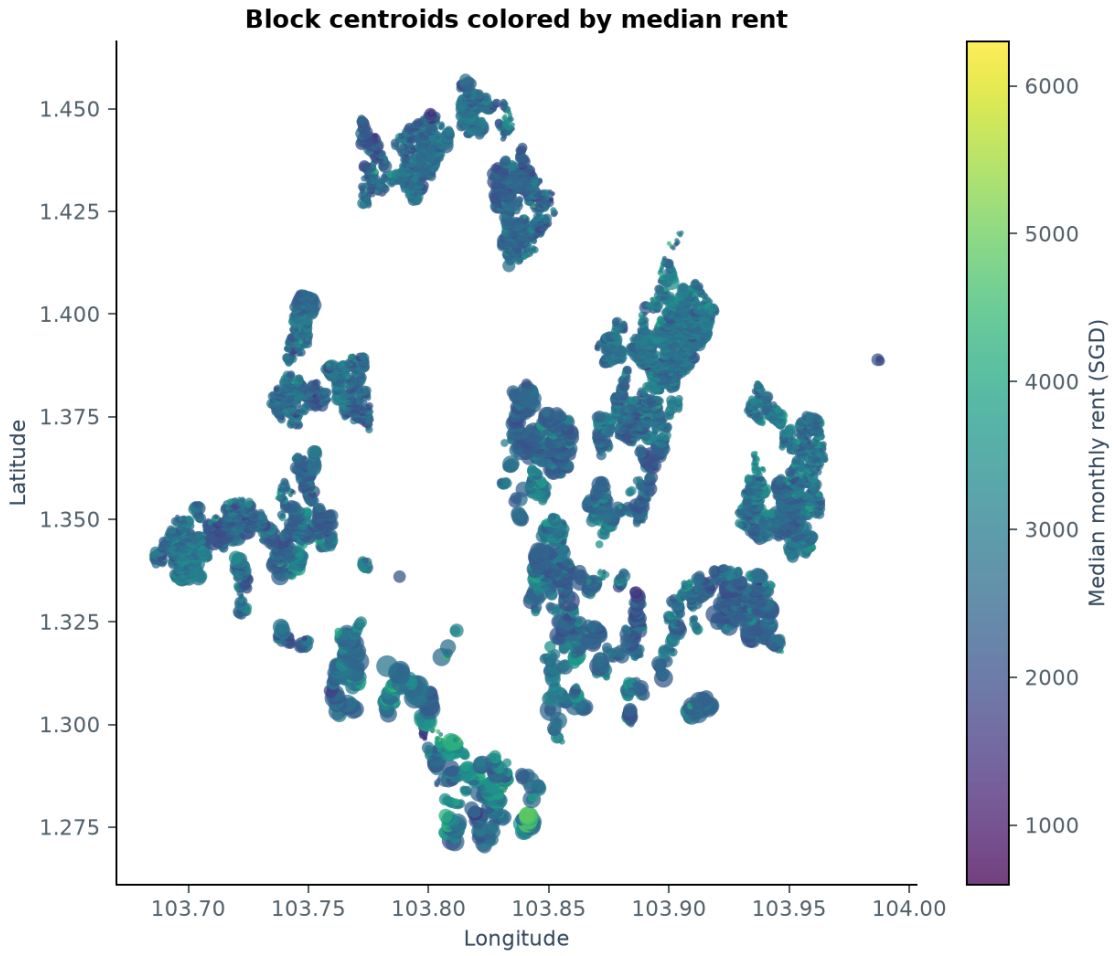
**100%**

train and test rows matched
a HDB block key

**9,464**

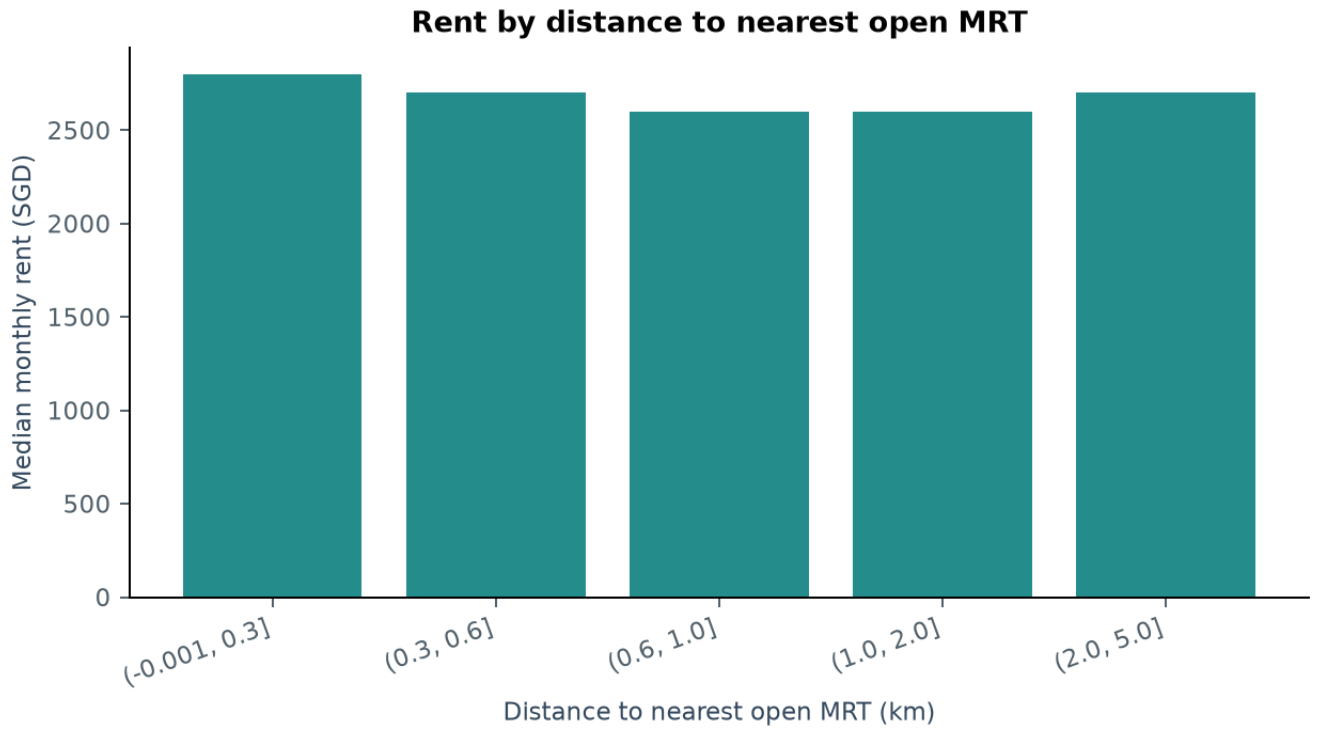
unique block keys

The map shows median rent at block centroids. Use location as a feature group and test its incremental value.




Source: data/auxiliary/sg-hdb-block.csv; map and join diagnostics: reports/eda/summary.json

# Raw distance associations are modest




- Nearest open MRT raw Spearman with rent: -0.086.
- Rent medians differ across distance bands ($2,800 near MRT; $2,600 at 0.6-1.0 km).
- Town, flat type, area and time are mixed in these unadjusted comparisons; do not read them as an MRT premium.

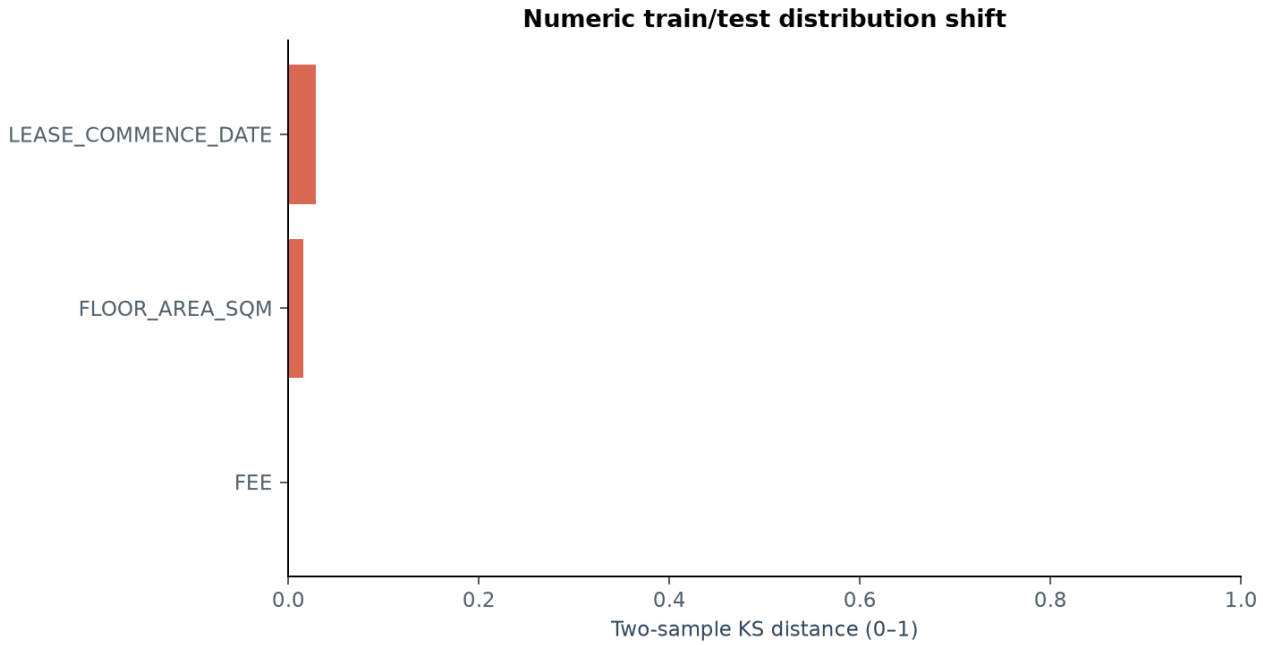
Source: reports/eda/figures/10_rent_by_open_mrt_distance.png; reports/eda/tables/location_feature_summary.csv

# Data quality decisions preserve prediction-row integrity

| | |
|---|---|
| **No missing values** | in the supplied train and test columns |
| **Constant fields** | FURNISHED = yes; FEE = 0 |
| **Label cleanup** | FLAT_TYPE 11 to 6 categories; STREET 1146 to 576 |
| **Repeated records** | 425 exact duplicate train rows; 5,508 exact duplicate test rows. Of 13,434 repeated feature groups, 13,191 have different rents. |
| **Policy** | Keep every test row in original order; do not deduplicate submission inputs |

Source: reports/eda/tables/data_quality.csv and categorical_normalization.csv

# The competition test set is a future-time test




- Train: 2021-01 to 2025-03.
- Test: 2025-03 to 2026-07.
- Most test dates fall after the final labeled month; no target labels exist to measure 2026 error.
- Sixteen test months are unseen in training (49,620 rows). Numeric KS distances are small for floor area and lease commencement year.

Source: reports/eda/summary.json; reports/eda/tables/train_test_drift.csv

# Auxiliary data creates useful features with timing constraints

| | | |
|---|---|---|
| **HDB blocks** | **9464** | coordinates, completion year, planning area |
| **MRT stations** | **243** | open and planned station points |
| **Malls / schools** | **113 / 337** | nearest-distance features |
| **COE / stocks** | **539 / 91,642** | monthly or daily time series |

Macro series are not merged in this EDA. Any later use must aggregate only information available by each approval date; same-month values can be published after the rental record. MRT status has no historical opening dates, so planned-station distances are current snapshots rather than time-valid features.

Source: data/auxiliary/*.csv; audit: reports/eda/tables/auxiliary_data_audit.csv

# Recent history improves the temporal baseline

| Validation | Fit window | Rows | MAE | RMSE |
|---|---|---|---|---|
| Random 20% | all dates | 30,000 | $567 | $668 |
| Jan 24-Mar 25 | all history | 44,163 | $787 | $888 |
| Jan 24-Mar 25 | previous 12m | 44,163 | $384 | $518 |
| Jan-Mar 25 | previous 12m | 8,924 | $345 | $491 |

On the same future window, restricting group medians to the previous 12 months reduces RMSE by 41.6% versus using all history. The random holdout mixes years and is not directly comparable. These are local validation scores; the competition metric is RMSE.

Source: reports/eda/tables/baseline_validation.csv; median predictor by TOWN x FLAT_TYPE

# Current decisions and open challenges

- Use chronological holdouts as the primary model-selection evidence; keep random split as a diagnostic comparison.
- Normalize text labels and preserve all test rows in source order.
- Use property, time and town with LightGBM/CatBoost as supervised benchmarks; add spatial features as a separate ablation.
- Treat nearest-place correlations as descriptive; compare adjusted model performance before attributing value.
- Remaining challenge: the labeled data stop in March 2025, while most competition inputs are later.

Evidence: reports/eda/summary.json and reports/eda/tables/*.csv

# Next steps | test auxiliary features

- Keep the current property + time + town models as fixed supervised benchmarks.
- Add HDB block attributes first, then MRT, mall and school distance features in separate ablations.
- Use date-aware joins for macro data and document assumptions where facility opening dates are unavailable.
- Compare every feature group on the same temporal split using RMSE, MAE and mean bias.

# Next steps | robust validation and final report

- Extend rolling-origin validation and compare full-history, recent-window and time-weighted training.
- Tune the strongest boosting candidates only after selecting feature groups.
- Check residuals by month, town and flat type, then report both improvements and failed additions.
- Generate 50,000 predictions in the original row order and validate the Id,Predicted submission schema.